

Using mindfulness to examine ISD agility

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Abstract. *As information systems development (ISD) organisations face increasing market demands, strategies such as agile development have been used to help address these demands. Now, as information systems (IS) research in agile development is gaining in theoretical rigour, this study explores the use of mindfulness as a theoretical framework to examine ISD agility. At its core, the mindfulness theoretical concept promotes a focus on 'continuous attention to detail' and 'vigilance to minimise errors and respond effectively to unexpected events'. Incorporating a sixteen month ongoing longitudinal case study of an ISD organisation, this research provides an insight into the practice of what we term 'doing' agile and 'being' agile, while also providing a number of theoretical contributions around the value of mindfulness for ISD agility.*

Keywords: agility, mindfulness, information systems development (ISD), case study

1. INTRODUCTION

Faced with increasing demands for information systems (IS), the software development industry has come under increasing pressure to produce higher quality systems more efficiently. With extensive formal planning/processes restricting traditional Waterfall methodologies, information systems development (ISD) organisations have increasingly shifted their focus towards agile methods (Lee & Xia, 2010). While these practitioner driven agile methods have been widely adopted, there has been little research community participation in the evolution of the agile methods themselves, which has led to a significant lack of empirical research in the domain (Conboy, 2009; Lee & Xia, 2010). However, in the last number of years, an increasing amount of empirical effort has been realised through the examination and exploration of agile ISD from numerous theoretical perspectives (cf. Fitzgerald *et al.*, 2006; Sharp & Robinson, 2006; Mathiassen & Vainio, 2007; Matook & Kautz, 2008). Furthering this empirical effort, this paper incorporates mindfulness as its theoretical lens, with the primary purpose of examining the ISD agility of an ISD organisation. While mindfulness focuses on similar goals as agility, it does so at an organisational level and has been utilised as a theoretical component in a number of domains, such as management science (Fiol & O'Connor, 2003; Levinthal & Rerup, 2006), organisational learning (Langer, 1997) and IS

(Swanson & Ramiller, 2004; Butler & Gray, 2006). However, building on existing links between agile ISD and mindfulness (cf. Matook & Kautz, 2008; Vidgen & Wang, 2009), this paper introduces the concept of 'being' agile vs. 'doing' agile. 'Doing' agile denotes the use of agile practices to address ISD process inefficiencies, whereas 'being' agile focuses on the underlying behaviours from which agility, or a lack thereof, is founded.

Examining the development of agile ISD and mindfulness, this paper explicitly outlines the theoretical links between the two domains. It is through these linkages that mindfulness is justified as an appropriate lens to examine the ISD agility of an organisation, with the implicit benefit of doing so also detailed. Incorporating a request from the chief executive officer (CEO) of an ISD organisation (to examine their ISD process), the case study, upon which this research is founded, is presented along with the data collection and analysis techniques utilised. The in-depth results, from the mindfulness examination of the organisation's ISD process, are then outlined, followed by the resulting implications for theory and practice.

2. THEORETICAL GROUNDING

2.1 Agile ISD

In the last number of years, there has been an increase in the volume of empirical case study research focusing on agile methods and their adoption/adaptation in organisational environments. These studies have examined agile teams in large organisations (cf. Fitzgerald *et al.*, 2006; Sharp & Robinson, 2006; Matook & Kautz, 2008; Conboy, 2009), in small firms (cf. Mathiassen & Vainio, 2007) and in comparative studies across organisational size, project type, project size and software development methodology (cf. Cao *et al.*, 2009; Vidgen & Wang, 2009). Furthermore, these studies have utilised a variety of theoretical perspectives, e.g. complex adaptive systems (CAS) (Vidgen & Wang, 2009), adaptive structuration (Cao *et al.*, 2009), dynamic capabilities (Mathiassen & Vainio, 2007), distributed cognition (Sharp & Robinson, 2006), agility (Conboy, 2009) and mindfulness (Matook & Kautz, 2008). Across the majority of these empirical research studies, one of the main contributions has been the development of a framework to facilitate a better understanding of agile practices and the management of agile teams. As a collective, these empirical research studies have strengthened the theoretical core of agile ISD, while presenting recommendations for further research in the area.

2.2 Organisational mindfulness

While there is an ongoing discussion about the definition of mindfulness (Kohls *et al.*, 2009), particularly due to the exponential increase in research investigating mindfulness and mindfulness-based interventions within psychology (cf. Chiesa & Serretti, 2011), the concept of mindfulness has emerged from Buddhist psychology (Kohls *et al.*, 2009) and meditation in practices (Hayes & Shenk, 2004). As a result of its origins, mindfulness is frequently

understood as a functional process that promotes a mental ability to focus direct and immediate attention on the present (present awareness), while taking a non-judgemental and non-evaluative stance on thoughts/experiences as they unfold (acceptance) (Hayes & Shenk, 2004; Kohls *et al.*, 2009). Despite these origins in meditative practices, evidence suggests that some non-meditative methods can be considered mindfulness techniques, where 'any method that increases attention to the present moment and an attitude of acceptance is a mindfulness method' (Hayes & Shenk, 2004, p. 250). As a result, this positions mindfulness as a process that promotes 'self-regulation of attention' and 'a posture of acceptance' in individuals (Hayes & Shenk, 2004, p. 249).

While the concept of mindfulness is most commonly associated with an individual's mindset, it has been introduced into organisational studies to distinguish between automatic/mindless and conscious/mindful information processing (cf. Walsh & Ungson, 1991). Furthermore, it is argued that organisational mindfulness provides a wider response portfolio, better process awareness and stronger accountability, which can lead to superior organisational performance (Weick *et al.*, 1999). Therefore, mindfulness can be regarded as an organisational/collective phenomenon, grounded in individual mindful behaviour that builds on organisational mechanisms (e.g. organisational routines) (Jordan *et al.*, 2009). In particular, it has been argued that 'collective mindfulness' is more than the sum of 'individual mindfulness' and ultimately relates to the distribution of decision-making within an organisation (empowered decision-makers) and the capabilities of a particular individual stakeholder (Butler & Gray, 2006; Vidgen & Wang, 2009). However, the challenge for organisations is how to promote mindfulness within their organisational boundary (cf. Jordan *et al.*, 2009).

Extant literature on managing with mindfulness typically focuses on internal interaction within an organisation (cf. Fiol & O'Connor, 2003). Furthermore, if mindfulness is viewed as a theoretical concept for understanding how organisations can achieve flexibility and reliability in their enactment of organisational routines, it is important that we understand the defining characteristics of the mindfulness concept. Building on the work of Langer (1989) in researching high reliability organisations, Weick *et al.* (1999) introduced five antecedent sub-processes that define mindfulness, namely: *reluctance to simplify interpretations*, *preoccupation with failure*, *sensitivity to operations*, *commitment to resilience* and *underspecification of structures (deference to expertise)*. Expanding on these definitions, Table 1 provides a description of mindfulness components and individual mindsets (cf. Weick *et al.*, 1999; Weick & Sutcliffe, 2007), aligned with their focus (cf. Fiol & O'Connor, 2003) and assessment (cf. Mu & Butler, 2009).

If mindfulness is viewed as a theoretical concept for understanding how organisations can achieve flexibility and reliability in their enactment of organisational routines, it is important that we understand the mindfulness concept from both the individual and organisational perspective, as presented in Table 1. While extant literature indicates that there are five central components that promote mindful behaviour within an organisational environment, it seems that the concept of mindfulness is somewhat paradoxical, due to the fact that to embrace this mindful behaviour within the organisation, organisational actors need to maintain a focus on the 'possibility of failure', 'recovery from failure' and 'failure avoidance'. Furthermore, there is

Table 1. Components of organisational mindfulness (after Weick *et al.*, 1999; Flol & O'Connor, 2003; Weick & Sutcliffe, 2007; Mu & Butler, 2009)

Mindfulness component	Focus on	Individual mindset	Organisational mindfulness assessment
Reluctance to simplify interpretations	Complexity to avoid failure	Uses 'cognitive simplifying processes' to make sense of and manage complexity	Questioning the way things are usually done here Challenging the status quo Appreciate scepticism here
Preoccupation with failure	The possibility of failure, especially if perceived as personal failure	May require a paradoxical focus on both success and failure to be effective	Prolonging analysis to better grasp nature of problems Taking even small mistakes seriously Seeing 'close calls' as mistakes Recognising consequence of team failures for the organization
Sensitivity to operations	Current operations and advocating change to avoid failure	Ensures 'situational awareness' of events as they occur in an effort to reduce uncertainty	Encouraged to share operations-related information Familiar with operations beyond our jobs Discuss operational issues in the team Management understands peoples' daily operations Staff here understand the nature of each others' jobs
Commitment to resilience	Recovery from failure	Includes a belief that 'bouncing back from setbacks must be a way of life'	Committed to solving any problem that arises People here can rely on each other Do not give up on solving a problem Developing peoples' skills/knowledge to increase their response repertoire
Deference to expertise	Appropriate expertise to avoid failure	Ensures an orientation towards experience to solve problems	Comfortable asking others with more experience for help To solve a problem, people ask someone with relevant experience Easy to obtain expert help with unexpected events Respect each others' professionalism

a need to balance these mindsets, e.g. previous research suggests that a continuous *reluctance to simplify interpretations* can lead to 'information overload' that negatively affects action, and a singular emphasis on failure (*preoccupation with failure*) may eliminate any sense of control and lead to 'learned helplessness' (cf. Fiol & O'Connor, 2003).

It is argued that organisational reliability can be enhanced (1) by changing existing policies/processes/procedures to eliminate those aspects that unwittingly promote mindlessness; or (2) by implementing policies/processes/procedures that strategists develop mindfully (Langer & Moldoveanu, 2000; Ndubisi, 2011). Either way, systems and policies/processes/procedures need to promote individual and collective mindfulness 'as a way of working marked by a focus on the present, attention to operational detail, willingness to consider alternative perspectives and an interest in investigating and understanding failures' (Ndubisi, 2011, p. 1). In a sense, mindfulness can be considered to be part of the third loop of triple loop learning that strives 'to create a setting where conscious collective mindfulness can be maintained' (Isaacs, 1993, p. 31).

Indeed, mindfulness is as much about the 'quality of attention' as it is about the 'conservation of attention' (Weick *et al.*, 1999; Rerup, 2005) and the 'enriched awareness' it generates facilitates an organisation to overcome unexpected events: 'notice more issues, process these issues with care, and detect and respond to early signs of trouble' (Rerup, 2005, p. 452). Therefore, mindfulness-based approaches 'focus on promoting highly situated human cognition as the solution to individual and organisational reliability problems' (Ndubisi, 2011, p. 2). As a result, mindfulness provides a succinct and compelling lens for viewing key aspects of reliability and is useful for confronting errors and failures (Butler & Gray, 2006; Ndubisi, 2011). Therefore, we argue that promoting collective mindfulness translates as promoting 'continuous attention to detail' and 'vigilance to minimise errors and respond effectively to unexpected events'. While this closely maps to the concept of agility, it does so at a higher level, thus providing the opportunity for further research and the potential to utilise mindfulness in examining ISD agility. The next section provides more detail into why it is used to examine ISD agility.

2.3 Examining ISD agility through mindfulness

By examining the literature around flexibility and leanness, Conboy (2009, p. 340) provides a definition of agility as follows: 'the continual readiness of an ISD method to rapidly or inherently create change, proactively or reactively embrace change, and learn from change while contributing to perceived customer value (economy, quality and simplicity), through its collective components and relationships with its environment'. Furthermore, as argued by Conboy (2009, p. 341) this definition 'provides an overarching interpretation of ISD agility', which is further represented as a taxonomy of ISD agility. This perspective on agility requires that 'an ISD method component' has the 'ability to handle change' and is 'continually ready'; however, it is argued that many agile practices 'do not contribute to agility as defined by the taxonomy' (Conboy, 2009, p. 342).

Indeed, Vidgen & Wang (2009), using CAS theory, develop an agile software development (ASD) organisation framework that identifies enablers and inhibitors of agility and the emergent capabilities of agile teams. In the discussion around their case study research findings, Vidgen & Wang (2009, p. 27) suggest that to 'optimise self-organisation' (a principle of CAS theory) a 'truly agile team' must embrace mindfulness. As part of this argument, Vidgen & Wang (2009, p. 27) suggest that team discipline should be 'self-regulated' and 'not simply imposed by managers' where the team can work with 'ease and satisfaction'; this is in direct contrast to mindlessness (which is defined by Butler & Gray, (2006, p. 215) as 'the mechanical use of cognitive and emotionally rigid, rule-based behaviours'). Vidgen & Wang (2009, p. 27) also conclude that 'in a self-managed team, a manager's role is more of a facilitator, creating an environment that fosters the emergence of self-organisation. In such an environment managers need to take a subtler approach than command and control and should nudge, remind, and reinforce agile behaviours through communication with team members'. A similar finding presented by Sharp & Robinson (2004, p. 9), based on their study of mature eXtreme Programming (XP) teams suggests that there is benefit for 'effective working' in a 'close-knit, informal setting'. Using a cognitive distribution theoretical framework, Sharp & Robinson (2004, p. 9) conclude that XP teams use 'a simple flow of information that is underpinned by shared understanding of the software under development and sufficient common ground to support effective communication'. Furthermore, Mathiassen & Vainio (2007, p. 534), examining dynamic capabilities in two small software firms, using a sense-and-respond framework as a lens, argue that small quick-responding teams coordinated by shared values and goals can 'anticipate individual customer needs and respond more quickly'.

Finally, Matook & Kautz (2008, p. 641), embracing the arguments of Butler & Gray (2006), propose that the characteristics of individual and collective mindfulness are also associated with ASD and suggest that the concept of mindfulness offers a 'theoretical foundation for understanding ASD'. Critiquing the fact that Butler & Gray (2006) did not provide a framework for studying the relationships between mindfulness and ASD (neither from a theoretical nor a practical perspective), Matook & Kautz (2008, p. 646) connect the characteristics of ASD and mindfulness through the Agile Manifesto, and illustrate the usefulness of this mapping with a case study, showing 'how mindful behaviour as part of an ASD approach may contribute to successful ISD'.

This paper argues that the agile prerequisites for agile ISD, as identified by Conboy (2009), should also be identified in software development team members, through examining their actions and perceptions, as opposed to examining the ISD method components themselves. We advocate that this departure can be facilitated through the promotion of collective mindfulness, where, at a conceptual level, mindfulness requires a 'continuous attention to detail' and 'vigilance to minimise errors and respond effectively to unexpected events'. Moreover, while ISD agility does recognise the behavioural aspect of software teams (Vidgen & Wang, 2009), we contend that the focus of that attention has been primarily on practices and methods. In contrast, mindfulness primarily focuses on behaviours but also recognises types of practices/routines albeit at a higher level (Butler & Gray, 2006; Jordan *et al.*, 2009). The examination of software development teams (and their behaviours) over methods, shifts the

focus away from particular ISD methods and associated practices towards ISD team capabilities, therefore moving the research agenda from 'doing' agile to 'being' agile. Furthermore, building on the mindfulness to agile ISD mapping we are cumulatively extending key previous research around mindfulness and ISD agility, while also attempting to address the lack of 'cumulative reflection' (Conboy, 2009, p. 331) by IS researchers on past ISD agility research.

3. RESEARCH DESIGN

Incorporating case studies as a research method is deemed appropriate in situations where 'little is known about a phenomenon', or 'current perspectives seem inadequate because they have little empirical substantiation', or 'they conflict with each other or common sense' (Eisenhardt, 1989, p. 548). Indeed, in the context of this research study, this 'freshness in perspective' is required on the topic despite the fact that agile methods themselves have received considerable attention in past research. From the perspective of this research study, the following description of the appropriateness of a case study to a particular type of research seems accurate: 'case research is particularly appropriate for certain types of problems: those in which research and theory are at their early, formative stages, and sticky, practice-based problems where the experiences of the actors are important and the context of action is critical' (Benbasat *et al.*, 1987, p. 369). Furthermore, an 'instrumental case study' research strategy was adopted for this study. This allowed the researchers to pursue a deep understanding of an issue as a primary focus, while the case itself was of secondary interest (as per Stake, 2000, p. 437). Therefore, to fulfil the objective of this research, studying a single instrumental case facilitated uncovering and teasing out the diverse organisational stories of those 'living the case' (Stake, 2000).

3.1 Case study

Emerging from a strong organisational necessity, the research objective was in line with the CEO's requirement for an external analysis of TexunaTech's ISD process and his belief that it could be more efficient. TexunaTech follow what could be described as a somewhat flexible waterfall method but acknowledge that it is inefficient. While considering themselves to be agile, they do not follow an agile method; the CEO thought that adopting an agile method may introduce increased efficiency in the organisation's ISD process. As a result, the primary motivation of this research was to examine the ISD agility of the organisation using mindfulness. Exploring the value of this theoretical lens provided a fresh perspective to understanding the organisation's ISD process and an insight into how true agility could be achieved in TexunaTech.

Having been setup just 10 years ago (now with over 40 employees and turnover in excess of £2 million), TexunaTech has established itself on the global market as a trusted service provider of web-based data management applications, serving a range of government, health-care and private sector organisations. TexunaTech values long-term relationships with its

clients, to meet and exceed their evolving expectations, which is evident in the long-term contracts they hold with their clients. Coupling geographical location with specific segments of the software development lifecycle, the organisations structure is defined as follows: (1) London (UK) – incorporates business analysis, project management and business development (although the role of Business Analyst (BA) and Project Manager (PM) is often performed by the same individual); (2) Cork (Ireland) – incorporates call centre operations, after sales service and first line support; and (3) Moscow (Russia) – incorporates software development, technology infrastructure maintenance and software testing.

3.2 Data gathering and data analysis

The 16-month exploratory research project was broken into two phases for the purposes of data gathering. The data gathering was conducted over a 7-month period (October 2009–April 2010) using a variety of techniques, as presented in Table 2. Phase 1 data gathering involved preliminary group-based interviews and discussions that provided a detailed background description of the organisation and its ISD process at a high level. It was during this phase that the research objective of mindful analysis was identified and aligned with the organisational request for an examination of the agility of the ISD process. Following on, phase 2 of the research strategy set out to further develop a deeper understanding of the organisation's mindfulness and ISD process. In particular, after phase 1, themes were identified from the preliminary group-based interviews/discussions. This provided the research team with the opportunity to focus their attentions for phase 2 data gathering. It was important that phase 2 data gathering was effective as the organisation made a significant commitment to the

Table 2. Case study research protocol (after Kelliher, 2005)

Research activity	Description
Objective	To identify the inefficiencies and overall agility of an ISD process
Approach	Case study (October 2009–January 2011)
Motivation	CEO's interest in achieving potential ISD process efficiencies through the adoption of agile practices
Case selection process	A software development organization where the CEO sought improvements in their ways of working
Case access	A unique openness to share information and a willingness to make personnel available for interview/workshops to the extent that operations were suspended for 3 days to enable the research
Instrument	The research team (three researchers) were the primary research instruments in the application of the data gathering techniques
Boundary device	ISD lifecycle
Data gathering techniques	On-site observations by research team, semi-structured interviews, round table discussions, intensive workshops with project team participants
Data analysis techniques	The components of mindfulness were utilised within the data reduction/analysis process to build a logical chain of evidence for the case

Table 3. Mindfulness perspective on the ISD process

Mindfulness components ISD process issues	Reluctance to simplify interpretations	Preoccupation with failure	Sensitivity to operations	Commitment to resilience	Deference to expertise
BA/PM helplessness	London –	London –			
Awaiting communication	Moscow –	Moscow –	London –		Moscow –
Buffer time erosion	London –	London –	London –	London +	
Pressure cooker development	Moscow –	Moscow –		Moscow +	

+ represents mindful behaviour; – represents mindless behaviour

research both financially and in terms of disruption to operations during the three workshops. Two of the workshops took place in Moscow and employees from the London and Cork offices were also made available for participation over a 3-day period. The third workshop took place in Cork, and several London employees were also made available for participation.

Furthermore, phase 2 data gathering workshops were organised around the projects, which defined the fabric of the ISD organisation to date. For the most part, these projects are revenue generating for external customers (predominantly long-term public sector contracts), with a growing number of internal customer projects initiated in more recent years – to introduce efficiencies through technology enhanced processing. Workshop participants were challenged to provide examples of inefficiencies in their day-to-day behaviour.

Analysing the data collected in phases 1 and 2, 'chains of evidence' were constructed to permit an 'explanation-building analytic strategy' (Lapointe & Rivard, 2005, p. 471). In order to discover patterns (behavioural issues and their impact on the ISD process) the coding of data involved categorising team participant's description of the ISD process against the five components of mindfulness (see Table 1). This categorisation was based on the sentiments expressed within participants' descriptions, extracted from the research team's analysis of the 25 hours of interview data and the 24 hours of workshop data. These data were further organised in 'conceptually distinct matrices' (against four key ISD process issues that emerged during analysis, namely: 'BA/PM helplessness', 'awaiting communication', 'buffer time erosion' and 'pressure cooker development') so as to build a logical chain of evidence (cf. Beaudry & Pinsonneault, 2005; Lapointe & Rivard, 2005). For the purposes of this paper a finalised output of our data analysis approach is presented in Table 3 (see section 4.2).

4. CASE ANALYSIS

Analysing the rich data gathered, using the mindfulness lens, a detailed picture of the organisation's mindfulness emerged. From this, a number of mindless behaviours were also identified that in turn impacted on the efficiency of ISD process. In particular, these mindless behaviours were found to be impacting on what we term the 'requirements documenting and use' phase of the ISD process. This name was used to represent requirements gathering

and the problems seen downstream with their subsequent use. It also differentiates the process examined from other aspects of the development process such as testing or version control.

4.1 Mindfulness analysis

Of particular note in the mindfulness analysis is TexunaTech's strong and positive *commitment to resilience*. All staff, across locations and job types, were able to give multiple examples of how the organisation was able to quickly adapt and respond to problems. At the end of a project, when time is running out and requirements have still not been clarified by their customer, TexunaTech is very effective and 'does whatever needs to be done' to ship the software on time. Although it is a very stressful time for those involved, everyone shows great flexibility in 'getting the job done'. This is a positive example of *commitment to resilience*, and is consistent with the 'recovery from failure' nature of this mindful behaviour. Nevertheless, further analysis found that this is a constant feature of most projects, and while TexunaTech have always been able to recover from these situations of incomplete requirements, it points to the existence of issues that place TexunaTech into recurring problematic states.

It became clear that fire fighting had become accepted as the norm by the London and Moscow teams. When asked for examples of efforts to prevent the reoccurrence of these problems, the teams accepted that it was not done and that they just moved on to the start of the next project. This issue was further qualified by analysing if errors were examined and addressed (*preoccupation with failure*). This highlighted the fact that while errors in coding were handled well, errors in process or behaviours were not addressed, leading to the continuous return to fire fighting at the end of projects.

Once it was identified that TexunaTech were poor with *preoccupation with failure*, other issues were raised by both teams concerning the other components of mindfulness; this was often seen where one team blamed another team for failings. With *sensitivity to operations*, the Moscow developers argued that the London BAs/PMs did not understand the implications of their behaviours and actions on the downstream ISD process. By London being over flexible in 'appeasing the customer', developers were left with poor and vague requirements that made their job extremely difficult. London, in turn, highlighted issues with *deference to expertise*, arguing that the Moscow developers did not take ownership of tasks, continually referring back to London (or internally to Moscow senior staff). While London accepted that they were the 'experts' and that they expected to be a source of information for the developers, they also felt that the developers should show empowerment and not need constant 'hand holding'; in effect, Moscow were taking *deference to experience* to the extreme. What became clear was that there was also an issue with *reluctance to simplify interpretations*. Moscow accepted that they did not challenge the status quo (the norm of London's 'customer appeasement'), while London ultimately accepted that they did not challenge the customer to provide requirements in a timely manner (and this ultimately gave rise to the 'customer appeasement' referred to by the developers).

4.2 A mindfulness perspective on the existing ISD process

Broadly outlining the mindfulness analysis, as presented above, facilitated highlighting the impact of the mindful/mindless behaviours on the organisation's ISD process, presented in the next section. From this, the mindful/mindless behaviours are explicitly discussed and directly linked to four key ISD issues within TexunaTech: (1) BA/PM helplessness; (2) awaiting communication; (3) buffer time erosion; and (4) pressure cooker development (see Table 3 where mindful and mindless behaviours are mapped by location against the mindfulness component and the ISD process where the behaviour was seen).

4.2.1 BA/PM helplessness

Initial findings from the mindfulness analysis of the organisation highlighted a failure in *reluctance to simplify interpretations* due to a focus on *preoccupation with failure* (which ultimately was problematic in itself as problems are not addressed), without any challenge to obvious reoccurring problems that have become the norm. Within TexunaTech, the BAs/PMs based in London are the first points of contact with their customers. Therefore, the BAs/PMs are responsible for gathering system requirements from these end users; however, managing these customers is a problem for the BAs/PMs, and as one described the situation: 'we often feel helpless when determining the requirements of the customer'.

This helplessness comes from the customers' lack of appreciation of the immediate nature of requirements specification as part of an efficient ISD process. However, the BAs/PMs' inability to challenge the status quo (*reluctance to simplify interpretations*) ensures this state of 'helplessness' constantly reoccurs. However, this issue is also compounded by the fact that TexunaTech's willingness to accommodate the customer is a unique selling point. Furthermore, the BAs/PMs' fear of making a mistake (e.g. not satisfying the customer or losing a client contract) is paradoxically exacerbating the helplessness, as they do not learn from their mistakes and close calls (*preoccupation with failure*). The resulting impact is felt in downstream ISD processes, as incomplete and late requirements creates issues for the software engineering part of the ISD process.

4.2.2 Awaiting communication

As already identified, London performs a number of mindless practices and push incomplete requirements downstream. This creates a situation where Moscow developers are *awaiting communication* for a number of requirements. An example provided of this situation was identified on a project where there were 12 vague/incomplete requirements remaining to be developed in a project release, with only 2 weeks development time remaining. The key point of this example is the fact that these requirements are considered high priority for the release by the BAs/PMs, but remain incomplete in terms of specification. Their incorporation into the existing system release will need to be researched by developers to avoid technical inconsistencies, and the collective development workload required to complete the release is greater

than the 2 weeks remaining. However, similar to London, Moscow displays evidence of an issue with a *reluctance to simplify interpretations* and particularly in challenging the status quo, which enables this practice to constantly reoccur. Moreover, Moscow display an overreliance on seeking guidance, something that senior developers in Moscow acknowledge as being excessive *deference to expertise*. This escalation procedure is a common occurrence that stresses the communication lines between the locations. While attention to technical details is always a priority, there are instances where operational inefficiencies are ignored (*preoccupation with failure*), leading to the problem where lessons are not learnt from operational/process errors. In addition, these issues are compounded by the lack of understanding from London of the effect of their procedures on the processes at the later stages of development (*sensitivity to operations*).

4.2.3 Buffer time erosion

During projects, Moscow personnel consistently flag vague/incomplete requirements that are high priority and nearing the end of a project release; this flags to the BAs/PMs that these requirements need to be specified immediately as they know a failure to do so will lead to a missed customer release. Labelled as *buffer time erosion* by the BAs/PMs, they then get the customer to provide a complete specification for the outstanding requirements and pass the details to the developers in the Moscow office. This further illustrates the BAs/PMs lack of understanding of their impact on the downstream ISD processes in the organisation (*sensitivity to operations*). Conversely, when problems with requirements are identified by Moscow, it is noted that London 'pulls out all the stops' and adapts, doing whatever is necessary to correct the problems (*commitment to resilience*). However, as this is a continuous reoccurring problem, questions must be asked as to why this has become the norm. This can be linked to BAs/PMs lack of learning from past mistakes (*preoccupation with failure*) and their failure to challenge the status quo (*reluctance to simplify interpretations*).

4.2.4 Pressure cooker development

Moscow personnel's understanding of the organisation is a true reflection of current operations; after all, they are the downstream ISD process actors that get impacted the most from the inefficient and ineffective 'requirements documenting and use' phase. Nonetheless, in Moscow, all personnel are encouraged to solve as many problems as possible in-house. For example, in an effort to complete the project release, additional developers may be seconded to the development efforts to ensure that the expected release date to the customer is not missed. It was reported by developers that this has an impact on progress made in other projects but is a necessary step to ensure customer satisfaction in terms of the delivery of expected functionality. Other issues are caused by this reassignment of developers but are beyond the scope of what is being specifically addressed in this paper, e.g. the late testing of functionality contained in the release. As a result, Moscow personnel (like London) also ensure they can deliver the project release to the customer on time through a strong *commitment to*

resilience by 'pulling out all the stops' and adapting team composition and creating ad hoc processes as necessary to get the product to the customer. However, as with London in *buffer time erosion*, questions must be asked as to why this has become the norm. By not challenging the status quo (*reluctance to simplify interpretations*), these problems have become the norm, and by continually passing over problems as opposed to addressing them (*preoccupation with failure*), there is nothing to stop these same problems occurring in future projects/releases.

4.3 Doing agile vs. being agile

As described in the preceding section, four main ISD issues were highlighted and mapped against the components of mindfulness (see Table 3) thereby uncovering the associated mindful/mindless behaviours. It must be stated that while these issues are unique to TexunaTech, they are similar to typical issues encountered by ISD organisations: vague and incomplete requirements (Albayrak *et al.*, 2009) leading to fire fighting (Blotner, 2002). In addition, *buffer time erosion* is similar to problems noted by others that risk identification is often an informal process that is performed badly (Ibbs & Kwak, 2000). Nonetheless, the mindfulness lens has afforded the opportunity to view these issues as manifestations of mindless behaviours. Based on current agile ISD thinking, the protocol is to prescribe agile practices that address ISD process inefficiencies ('doing' agile). Yet, by utilising mindfulness as a lens it is possible to prescribe solutions that shift the focus more towards the mindless behaviours themselves, which are the root cause of the ISD process inefficiencies ('being' agile). To further develop this analysis, these two potential perspectives are outlined and supported for TexunaTech in the next section.

4.3.1 Doing agile in TexunaTech

Agile methods (such as XP) recommend that requirements should be written in the form of user stories which describe simply what the customer wants (Cohn, 2004). This simplicity (Beck & Fowler, 2001) enables customers to better define their requirements, mitigating the issue for what is defined in TexunaTech as *BA/PM helplessness*, where there are difficulties in getting information from, and collaboration with, the customer. It should also help to lessen the problems further down the development process in what is defined in TexunaTech as *awaiting communication*, where requirements are vague or incomplete. With clear, simplified, requirements there should be less likelihood of developers discovering vague and incomplete requirements at the later stages of the development process when it is too late to deal effectively with them (*buffer time erosion* in TexunaTech). Recommending that the customer defines user stories does not necessarily ensure that it will happen all the time. As the customer determines the initial requirements through user stories, they should be collected into a product backlog list (Cohn, 2004; Fitzgerald *et al.*, 2006). A core element of the product backlog list is that all user stories must be prioritised by the customer (Baskerville & Pries-Heje, 2004; Cohn, 2004). As a result of implementing these practices, a more efficient and agile process should theoretically be in place and alleviate any *pressure cooker development* ISD process issues.

4.3.2 Being agile in TexunaTech

By adopting agile practices, such as those above, an organisation can consider itself to be following an agile methodology, i.e. 'doing agile'. However, while addressing the ISD process issues within TexunaTech, these practices fail to address the core behavioural issues identified in the mindfulness analysis. Thus, TexunaTech will not become agile in the true sense of the word. To move from 'doing' agile to 'being' agile, TexunaTech must also focus on the mindless behavioural issues that are the root cause of the ISD process issues outlined. Otherwise, the practices implemented are more akin to a short-term solution to symptoms of a deeper problem. While retrospectives are commonly used in agile (Augustine *et al.*, 2005), it is typically a self-reflection process that aims to improve practices and work methods (Salo, 2004). It is argued herein that the emphasis of retrospectives needs to move from process-based to (mindful) behaviour-based reflection. During the data gathering phase of this research, a number of workshops were undertaken. From TexunaTech's perspective, these workshops were independently moderated and focused on problem analysis within the ISD process. These workshops could be regarded as retrospectives that promoted behaviour-based reflection. For instance, the independent moderation allowed (and at times pushed) TexunaTech employees to explicitly challenge the status quo (*reluctance to simplify interpretations*), while the overall exercise directly focused on their *preoccupation with failure* abilities. Furthermore, while the workshops brought an element of mobility to TexunaTech employees (BAs/PMs and developers sharing face-to-face experiences for the first time), this was an example of a further behaviour changing solution. This employee mobility and co-location (for a 3-day period) facilitated a better understanding and awareness of the day-to-day operations (*sensitivity to operations*) within the organisation and ownership was taken by Moscow developers in particular to alter their behaviours to alleviate TexunaTech's issue with excessive *deference to expertise*.

5. CONCLUSIONS: MINDFULNESS ROUTINES AND BEING AGILE

As presented earlier in the paper, the definition of agility proposed by Conboy (2009) is said to provide an overarching interpretation of ISD agility and requires that an ISD method component has the ability to handle change and is continually ready; however, it is also argued by Conboy that many agile practices do not contribute to agility. Furthermore, restating our point of departure for this paper, we proposed that the presence of the prerequisites for ISD agility, as identified by Conboy (2009), should be identified in software development team members, through examining their actions and perceptions (behaviours), as opposed to only examining the ISD method components (agile practices) themselves.

Based on the case analysis of the ISD process presented in the previous sections, we argue that this departure can be facilitated through the promotion of organisational mindfulness, which at a conceptual level mindfulness, requires a 'continuous attention to detail' and 'vigilance to minimise errors and respond effectively to unexpected events'. Therefore,

ISD Agility	Mindfulness
ability to handle change	vigilance to minimise errors and respond effectively to unexpected events
continually ready	continuous attention to detail
Agile Practices	
- user stories - product backlog lists - prioritisation	retrospectives
'Doing' Agile	'Being' Agile
Mindfulness Routines	

Figure 1. 'Doing' and 'Being' agile through mindfulness routines

mindfulness strives to promote the prerequisites for ISD agility, where agile practices themselves may fall short. This examination of software development teams over methods, shifts the focus away from particular ISD methods and associated practices and towards ISD team capabilities (mindful behaviours); this therefore moves the research agenda from 'doing' agile to 'being' agile, as illustrated in Figure 1.

Furthermore, building on the mindfulness/ISD agility mapping, we are cumulatively building on key previous research around mindfulness and agile ISD (cf. Matook & Kautz, 2008), and are going some way to addressing the lack of 'cumulative reflection' (Conboy, 2009, p. 331) by IS researchers on ISD agility research. To further this 'doing' agile and 'being' agile philosophy, we can return to the work of Vidgen & Wang (2009) who suggested that a truly agile team must embrace mindfulness and team discipline should be self-regulated. While mindfulness is an individual activity, and the state of an individual's mindfulness is manifested in an individual's sense making/understanding of the organisational environment, to promote and maintain organisational mindfulness and 'reflection-in-action' we conclude that organisations need to create an environment to introduce 'mindfulness routines', and behaviour-based reflection (retrospectives) provide a mechanism for doing so.

From the perspective of our research work, we visualise a mindfulness routine as an organisational routine (e.g. the 'doing' agile practice of user stories) with a self-assessment dimension promoting reflection-in-action (e.g. the 'being' agile practice of retrospectives). Therefore, actors performing an organisational routine have the ability to provide immediate input in a structured manner on the efficiency or otherwise of the routine. This suggests that ongoing improvements become a feature of the organisational routine itself (e.g. the 'doing' agile practice of user stories or prioritisation). In adopting the perspective above, the organisational routine would become a mindfulness routine, where we define a mindfulness routine as: *'new information from multiple perspectives, gathered through self-assessment/reflection, to promote continuous creation and refinement of the performative aspect of an organisational routine'*. Therefore, we argue that mindfulness does not exclude or oppose the idea of routines.

In fact it may often build upon 'routinised action' in certain organisational environments (Levinthal & Rerup, 2006; Jordan *et al.*, 2009), especially if the routines are perceived as being 'interactive routines' or 'mindfulness in interaction' based on 'activities and routines that explicitly aim at providing opportunities to question expectations and behavioural routines and to evoke awareness context in interaction' (Jordan *et al.*, 2009, p. 468).

As a result, this positions mindfulness as a process that promotes 'self-regulation of attention' and 'a posture of acceptance' in individuals (Hayes & Shenk, 2004, p. 249). Again, this solidifies our 'doing' agile and 'being' agile philosophy, where organisational actors need to maintain a continuous focus and vigilance on the 'possibility of failure', 'recovery from failure' and 'failure avoidance', during their performance of an organisational routine (e.g. the 'requirements documenting and use' phase of the ISD process).

For future ISD research, the mindfulness theoretical lens has significant potential. For example, rather than solely concentrating on the ISD process, or the potential of agile methods to enhance the process, mindfulness enables researchers to determine what team behavioural issues are impacting on the ISD method components of the ISD process, and then to return to the ISD process to determine a solution. This approach could be considered a hermeneutic approach, specifically the hermeneutic cycle, which is the dialectic between the understanding of an action as a whole and the understanding of the parts of the action (Surendra, 2008). It is argued herein, that the mindfulness lens provides researchers with a theoretical tool to better examine the ISD process, and to determine the suitability of solutions (as agile can be for TexunaTech in this research).

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